

psychologists often distinguish between **subjective probabilities**, which are influenced by characteristics of the probability estimator, and *objective* probabilities, which are not. Of course, in many real-life circumstances there may be no objective probabilities available.

■ COGNITIVE ILLUSIONS IN DECISION MAKING

How do people gather the information they will use to make a decision? Often the information comes from their own memories. Students choosing a major, for instance, may think back to their experiences in different courses or to things they have heard older students say about their experiences in different majors. Once information is gathered, the decision maker must decide on the importance and/or relevance of each piece of information. If you don't care about becoming a biologist, for instance, the information on the biology department's course offerings may not seem very important to you. The way people gather and assess the relevance of different pieces of information is the topic of this section.

Research on people's decision-making skills and styles has consistently demonstrated the existence of certain systematic and common **biases**, ways of thinking that lead to systematic errors. Typically, the biases are understandable and often justifiable ways of thinking under most conditions but can lead to error when misapplied. These systematic biases have been labeled **cognitive illusions** (von Winterfeldt & Edwards, 1986b). The term itself is meant to invoke the analogy to perceptual illusions: errors of cognition that come about for understandable reasons and that provide information relevant to understanding normal functioning. We can and do consider these illusions "errors," in the sense that one's percept does not correspond with what's really there. However, these illusions are not used as evidence that the whole perceptual system is faulty and unreliable. Rather, illusions (perceptions under certain specific conditions) tell us something about the way the perceptual system works generally—what cues are attended to, how they are interpreted, and so forth.

In a similar way, errors in decision making tell us something about the ways people gather, sort, and integrate the information they use for making a choice. The cognitive illusions described next also give us information on when unaided human decision making is likely to be optimal and when it is not. Finally, these descriptions can help us design and implement educational programs or interventions to improve the quality of decisions and plans people make.

Just what is a cognitive illusion? Von Winterfeldt and Edwards (1986b) specified that something counts as a cognitive illusion only if there is a "correct" way of answering a question or making a decision, there is also an intuitive estimate or decision, and there is a discrepancy between the two that always

goes in the same direction. Answers that randomly fluctuate around the correct value, then, do not count as illusions.

Availability

Consider the problems in Box 13-1, and give your first intuitive response to each before reading further. Tversky and Kahneman (1973) presented problems such as these to undergraduate students. The general findings were that people's intuitions were systematically wrong. In Problem 1, for instance, the letter *L* occurs more frequently in the third position than in the initial position. In Problems 2 and 3, the A and B options have the same number (of committees in the former case, and of paths in the latter).

Box 13-1 Problems Demonstrating Availability

1. Consider the letter *L*. In the English language, is this letter more likely to appear in the first position of a word or the third position of a word? Give your intuition or "gut reaction."
2. Ten students from a nearby college have indicated a willingness to serve on a curriculum committee. Their names are Ann, Bob, Dan, Elizabeth, Gary, Heidi, Jennifer, Laura, Terri, and Valerie.
 - a. The dean wants to form a two-person committee. What is your estimate of the number of distinct committees that could be formed? (Don't use formulas; just respond intuitively.)
 - b. The dean wants to form an eight-person committee. What is your estimate of the number of distinct committees that could be formed? (Don't use formulas; just respond intuitively.)
3. Consider the two structures shown below:

A	B
x x	x x x x x x x x
x x	x x x x x x x x
x x	x x x x x x x x
x x	
x x	
x x	
x x	
x x	
x x	

A *path* in a structure is a line that connects one "x" from each row, starting with the top row and finishing at the bottom row. How many paths do you think each structure has? (Again, give an intuitive estimate.)

SOURCE: Adapted from Tversky and Kahneman (1973, pp. 212–214).

What accounts for the errors? Tversky and Kahneman (1973) argued that when faced with the task of estimating probability, frequency, or numerosity, people rely on shortcuts or rules of thumb, known as **heuristics**, to help make these judgments easier. One such heuristic is known as the **availability heuristic**—“assessing the ease with which the relevant mental operation of retrieval, construction, or association can be carried out” (p. 208). In other words, instances (for example, particular words, particular committees, or particular paths) that are more easily thought of, remembered, or computed stand out more in one’s mind. Those instances are particularly salient and hence are deemed to be more frequent or probable.

In Problem 1, it turns out to be easier to think of words that begin with *l* (such as *lawn*, *leftover*, and *licorice*) than to think of words that have *l* as the third letter (*bell*, *wall*, *ill*). The reason for this may stem from the way our lexicons, or “mental dictionaries,” are organized or with how we learn or practice words—alphabetically by the first letter. As with paper or electronic dictionaries, it’s relatively easier to search for words by initial letter than by “interior” letters.

In Problem 2, the appropriate formula for determining the number of distinct committees that can be formed is $10!/\{(x!)([10-x]!)\}$, where x is the size of the committee. Notice that for $x = 2$, $10 - x = 8$, and that for $x = 8$, $[10 - x] = 2$, implying there should be an equal number of two-person committees and eight-person committees (namely, 45). Tversky and Kahneman (1973) argued that two-person committees are more distinct. There are five two-person committees with no overlap in membership, but any two eight-person committees will have at least some overlap. Distinctiveness makes different committees easier to think of. Therefore, two-person committees are more available (because they are more distinctive) and hence deemed more numerous. You can easily see, however, that two-person and eight-person committees must be equally numerous. Consider that every two-person committee defines an eight-person noncommittee, and vice versa.

The same kind of analysis applies to Problem 3. The number of paths in either structure is given by the formula xy , where x is the number of x ’s in a row and y is the number of rows. The number of paths in structure A, then, is $8^3 = 512$. The number of paths in structure B is 2^9 , also equal to 512. Again, though, it is easier to see more nonoverlapping paths in A than in B; different paths in A are less confusable than different paths in B. Paths in A are shorter and therefore easier to visualize than those in B. The ease of visualization makes paths more available and hence deemed more numerous in A than in B.

Everyday analogs that involve the use of the availability heuristic have also been reported. Ross and Sicoly (1979), for instance, surveyed 37 married couples (husbands and wives separately and independently) about the estimated extent of their responsibility for various household activities, such as making breakfast, shopping for groceries, and caring for children. Husbands and wives

both were more likely to say they had greater responsibility than did their spouse for 16 of the 20 activities. Moreover, when asked to give examples of their own and their spouse's contributions to each activity, each spouse listed more of her or his own activities than activities of her or his spouse.

Ross and Sicoly (1979) explained these findings in terms of the availability heuristic. Our own efforts and behaviors are more apparent and available to us than are the efforts and behaviors of others. After all, we are certain to be present when we perform an action, but we may or may not be when a friend or spouse does. Our own thoughts and plans are important to us, and we may be formulating them just at the time when other people do or say something, thus distracting us from their contributions. In general, what we do, think, say, or intend is more accessible to us than to anyone else and also more accessible than anyone else's deeds, thoughts, words, or intentions. Small wonder, then, that in joint ventures each partner often feels she or he shoulders a greater share of the burden.

Availability can be both an efficient and effective heuristic. If we can be sure that ease of constructing or calling instances to mind is unbiased, then it may be the best, or even only, tool to use when judging frequency or probability. If you are trying to decide which course you typically do more papers for, psychology or philosophy, it probably is fair to judge the frequency of papers by trying to recall specific paper assignments for each course. In this case, there is probably no particular reason to believe psychology papers are more memorable than philosophy papers. If there is (for example, you took philosophy three years ago but psychology this semester), then the comparison is probably not fair.

However, if you are trying to decide which occurs more often, hours you spend working on a group project or hours someone else spends working on the same project, using availability to judge may be unfair. You have been there whenever you have worked, but you may not have been there all the times when other group members have worked. And even if you had been there, you probably would have been paying more attention to your own work and planning than to your partners' work and planning. Thus examples of your own work are likely to be more memorable and more available to you than examples of anyone else's work.

The point of demonstrating the availability heuristic, then, is not to warn you away from its use. Instead, as with all other heuristics, the idea is to suggest you think carefully first about whether the range of examples you are drawing from is equally accessible.

Representativeness

Two students, Linda and Joe, are having a boring Saturday afternoon in the student union. For lack of something better to do, they each begin flipping a quarter, keeping track of the way it lands over time. Then they compare results. Linda